Physical Chemistry (I)

Lecture 19

Solid-Liquid Equilibria

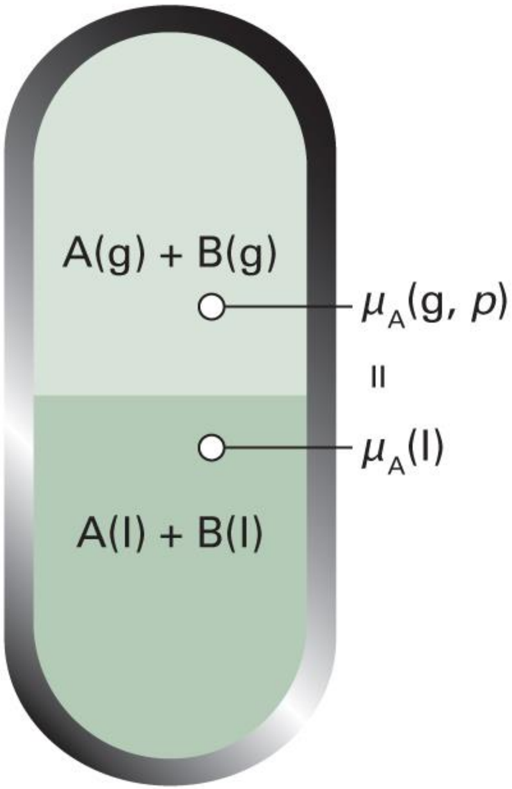
In the Last Class

- Phase diagrams of ideal solutions
  - Compositions of liquid and gas phases
  - Composition-pressure diagrams
  - Composition-temperature diagrams

- Phase diagrams of real solutions
  - Azeotropes
  - Phase separation

Last class

Liquid-Vapor Equilibrium



- Constraints

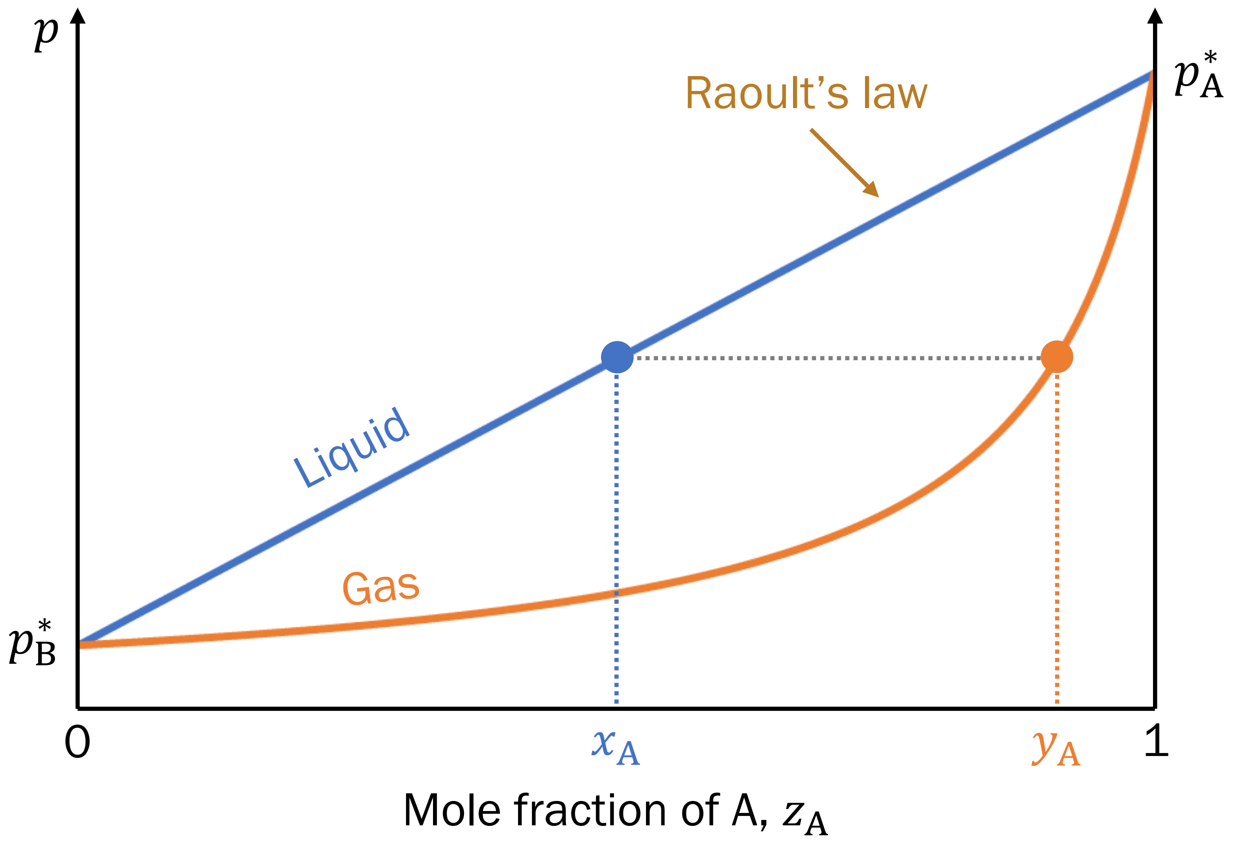
$$\mu_A(g) = \mu_A(l)$$
$$\mu_B(g) = \mu_B(l)$$

- May have $x_A(g) \neq x_A(l)$

(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 12th ed., Figure 5A.10)

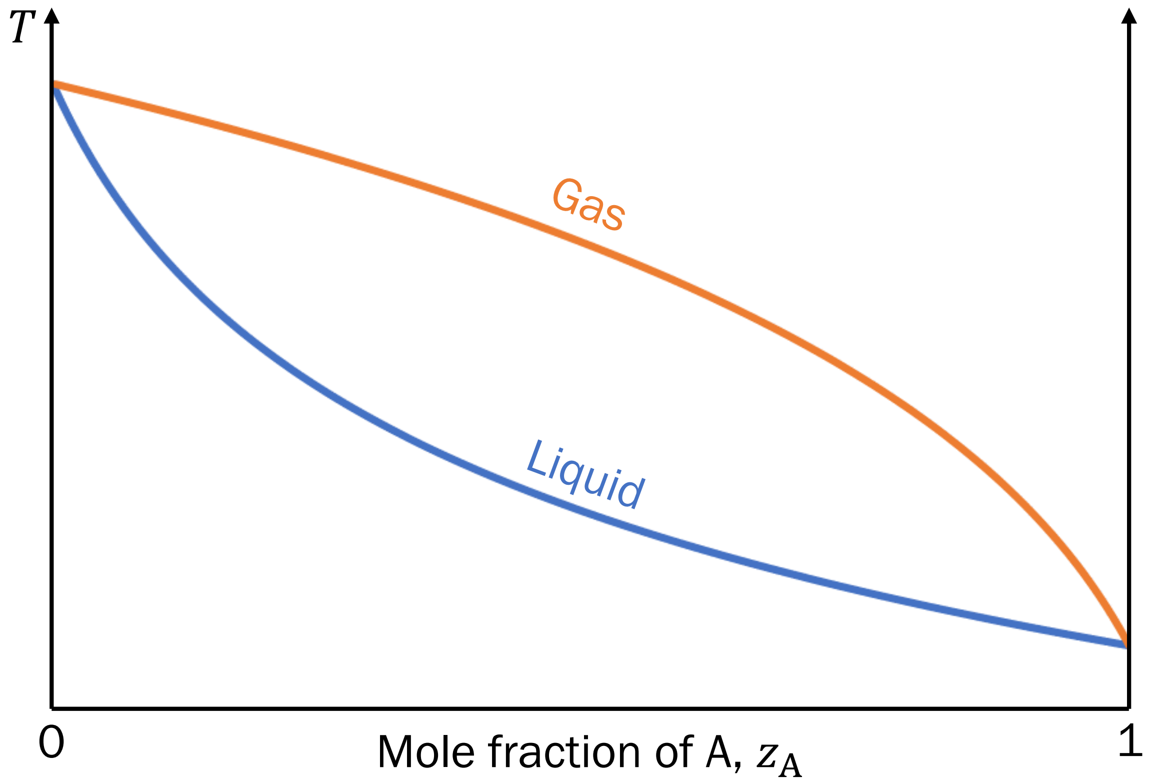
Last class

Composition-Pressure



Composition-Temperature

$p_A^* > p_B^*$ (A is more volatile)



Real Solutions

$$\xi = \frac{z}{k_\mathrm{B}T}\left(w_\mathrm{AB} - \frac{w_\mathrm{AA} + w_\mathrm{BB}}{2}\right) \neq 0$$ (BW model)

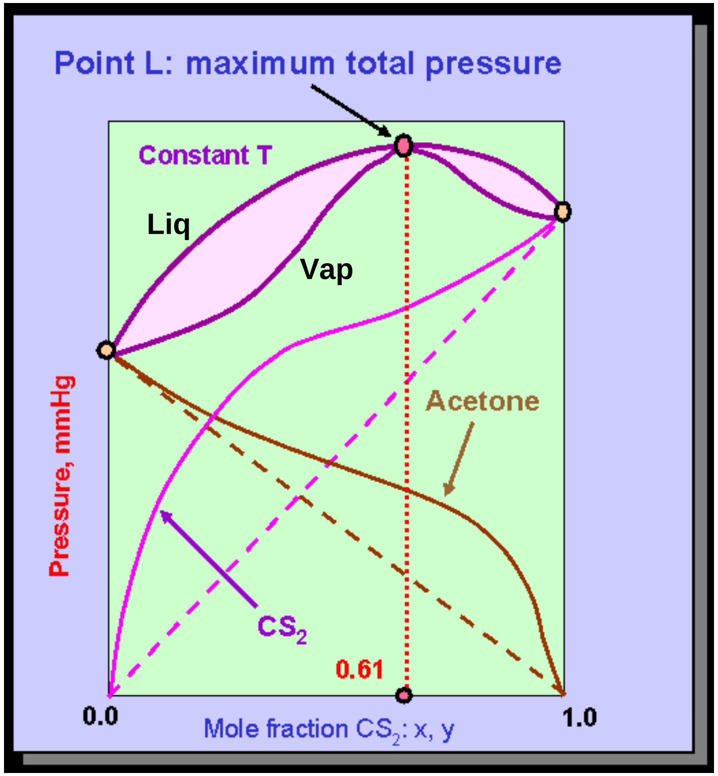
- $\xi > 0$: AB interactions are weaker
- $\xi < 0$: AB interactions are stronger

1. Change in liquid-gas equilibrium: azeotrope
2. Change in liquid-liquid equilibrium: phase separation

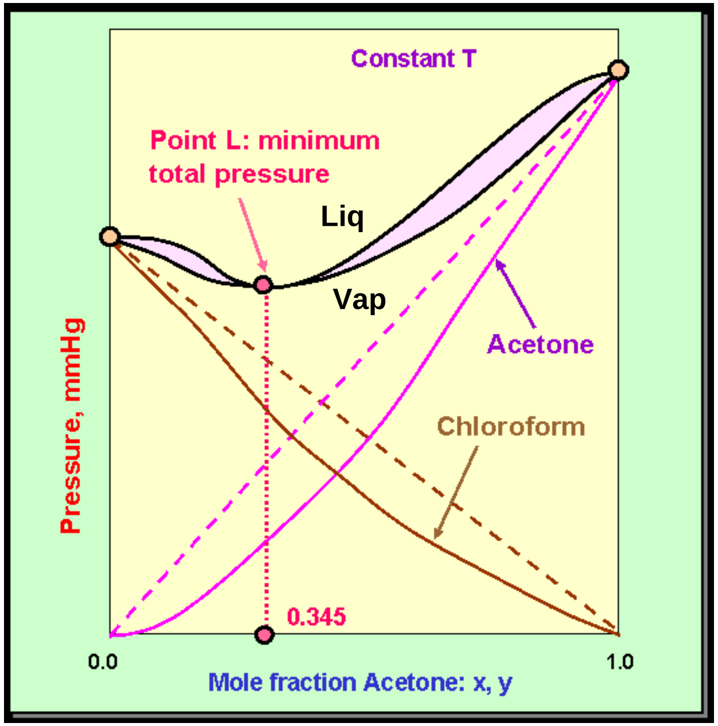
Last class

Composition-Pressure

$\xi > 0$



$\xi < 0$

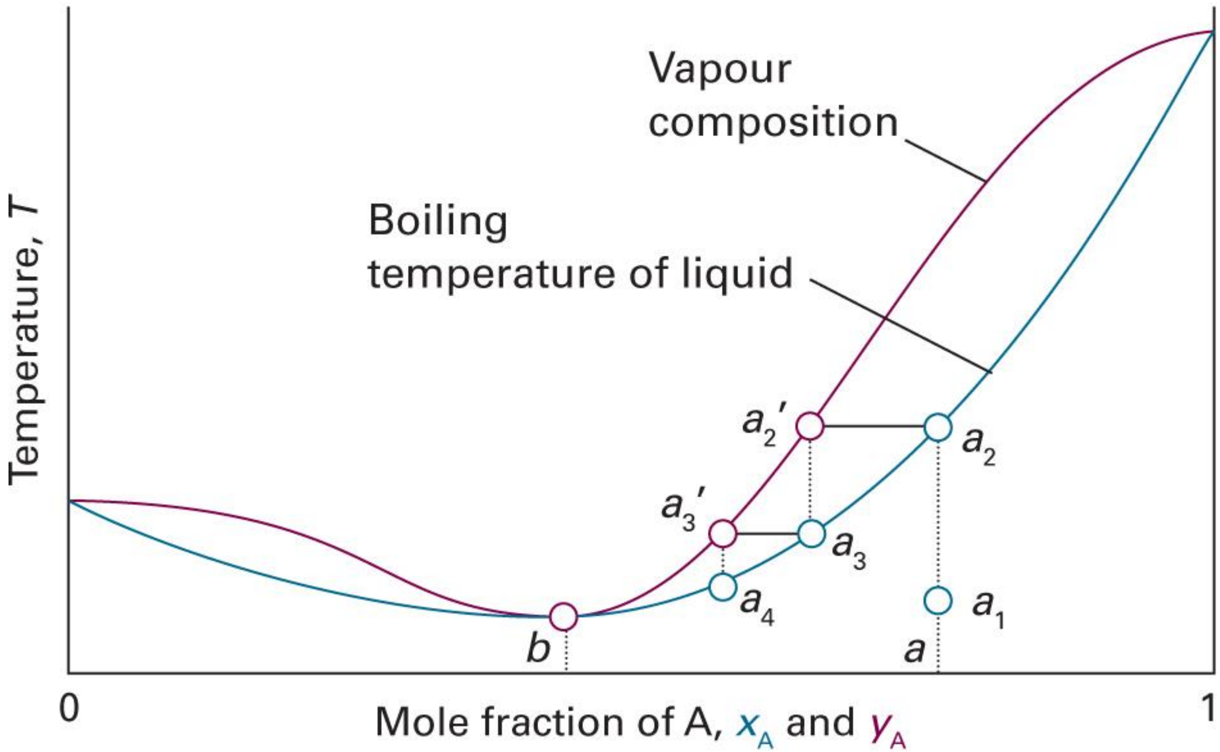


(Image: http://www.separationprocesses.com/Distillation/DT_Chp01f.htm)

Last class

Low-Boiling Azeotrope

$(\xi > 0)$

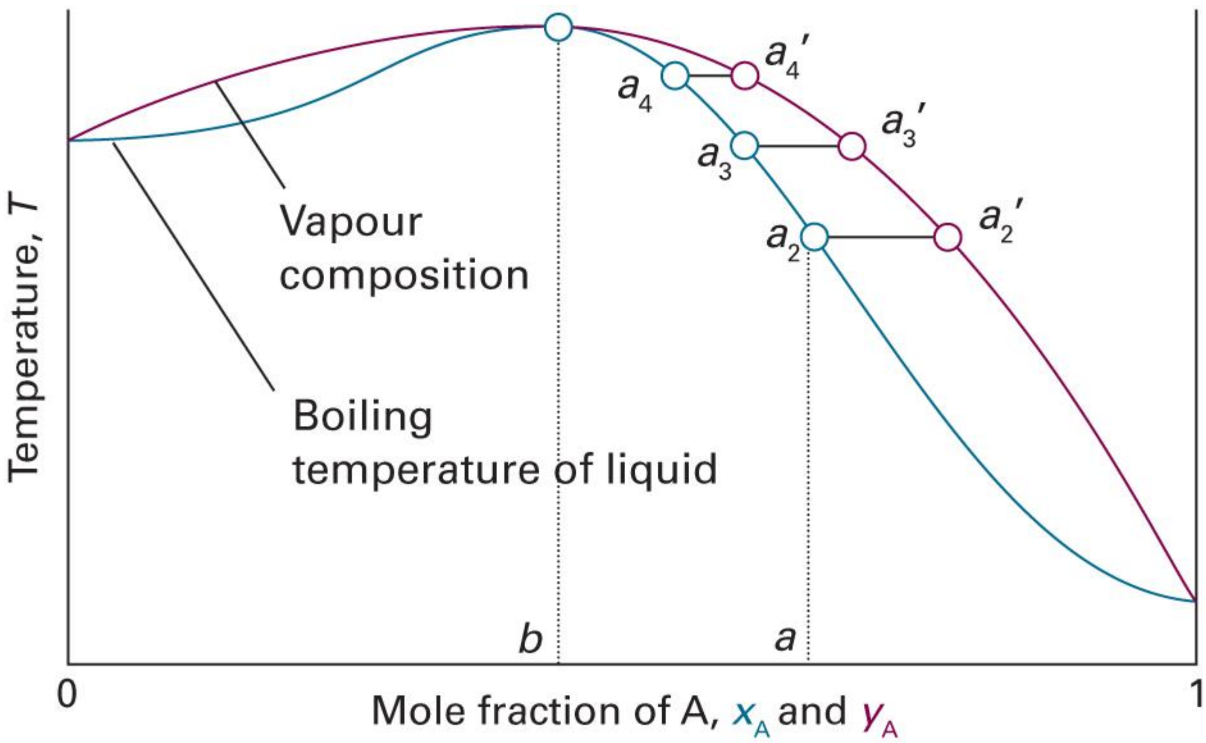


(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5C.12)

Last class

High-Boiling Azeotrope

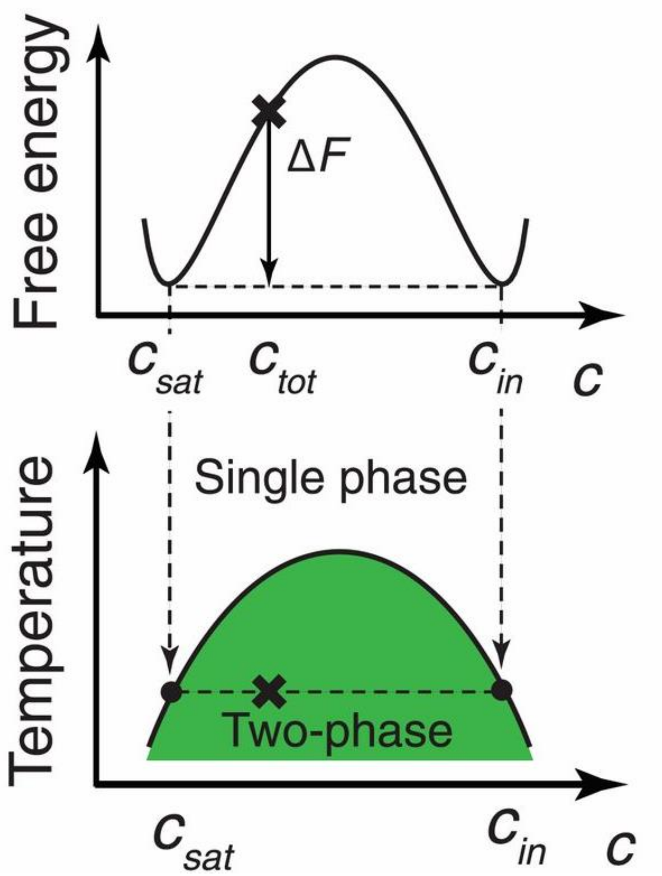
$(\xi < 0)$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5C.11)

Last class

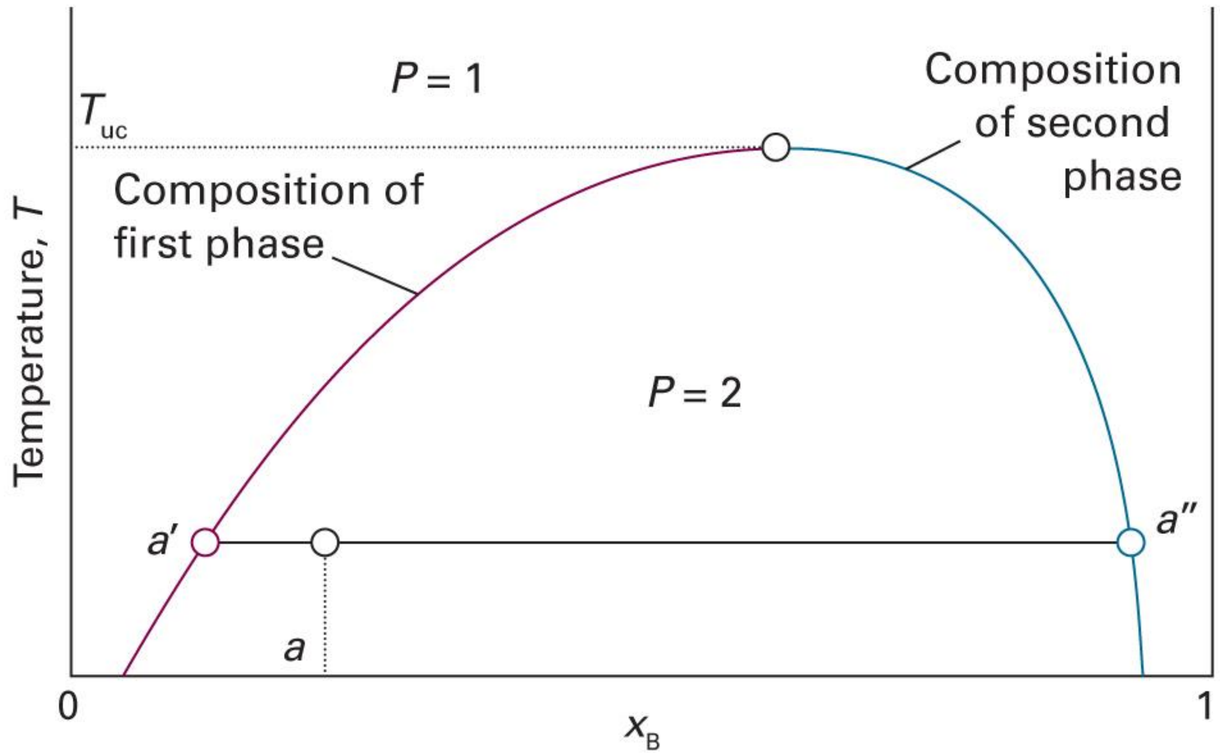
Phase Separation



(Image: Shin and Brangwynne, ***Science* 357**: eaaf4382 (2017), Figure 3)

Last class

Upper Critical Solution Temperature

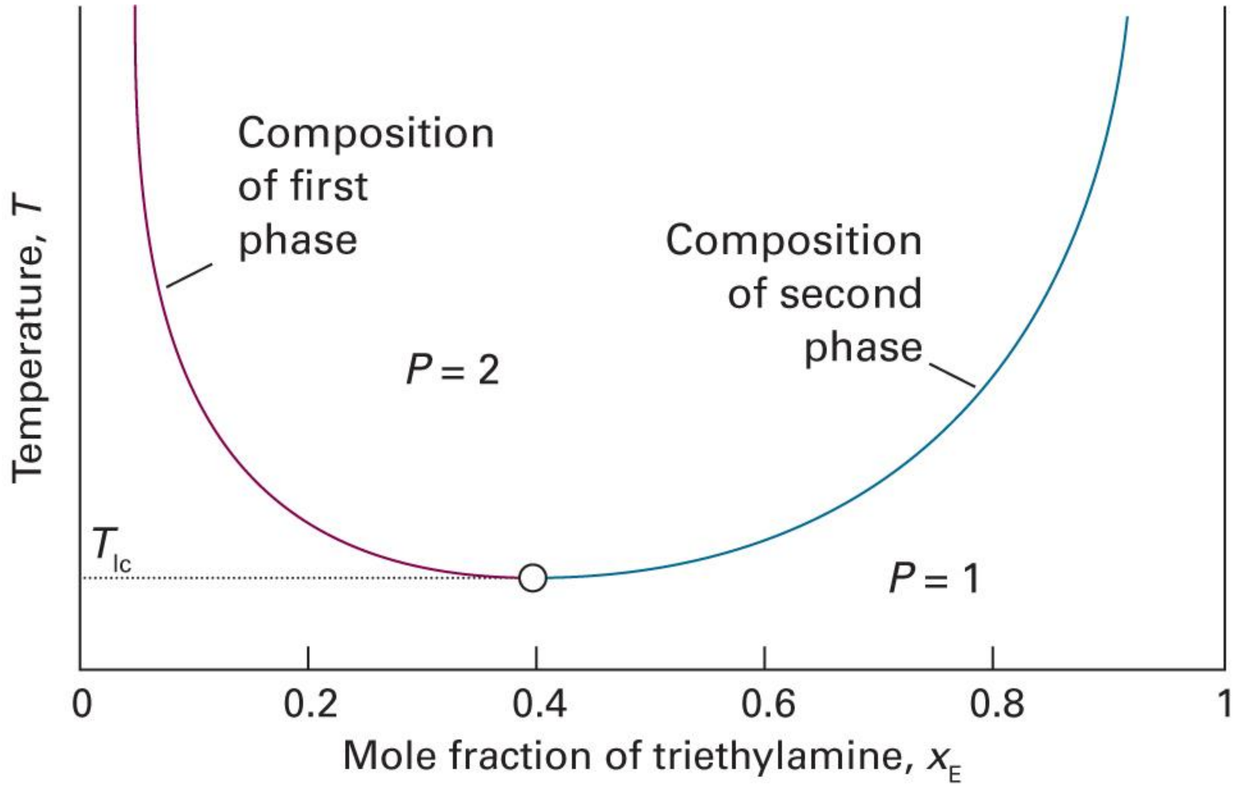


(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5C.14)

Last class

Lower Critical Solution Temperature

water + triethylamine



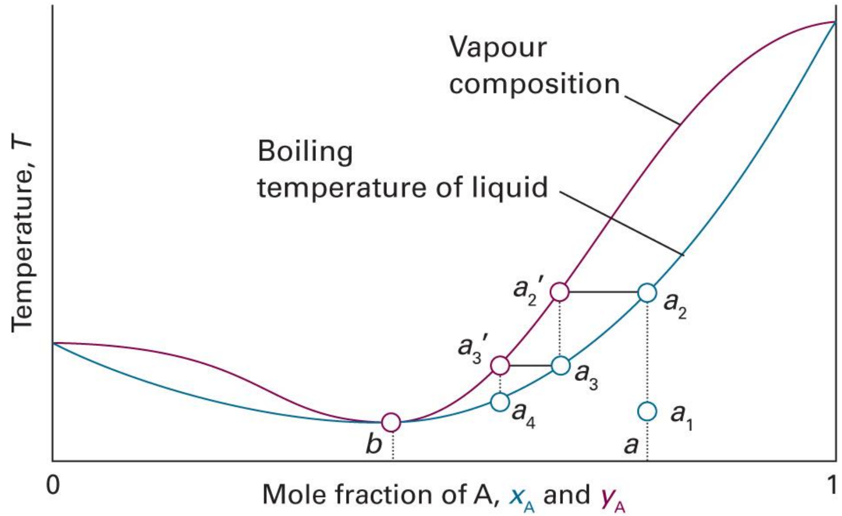
Two components form weak complexes at low T

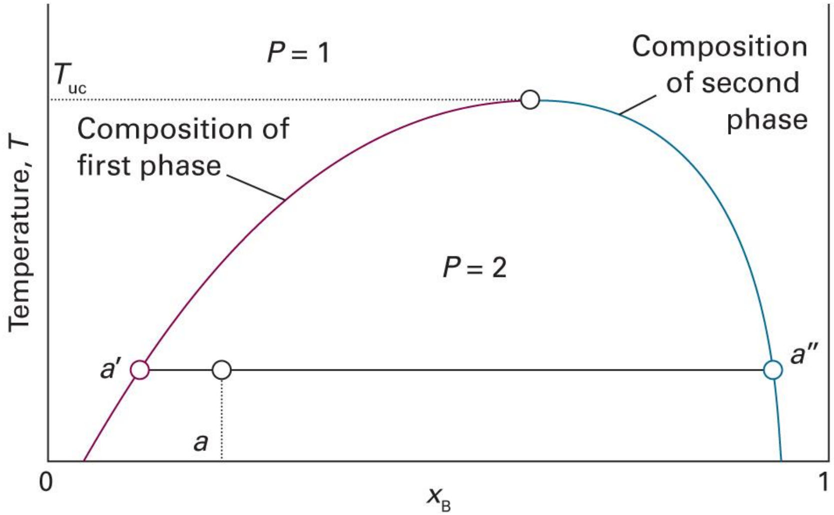
(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 12th ed., Figure 5C.19)

Last class

Azeotrope + Phase Separation

(when $\xi \gg 0$)

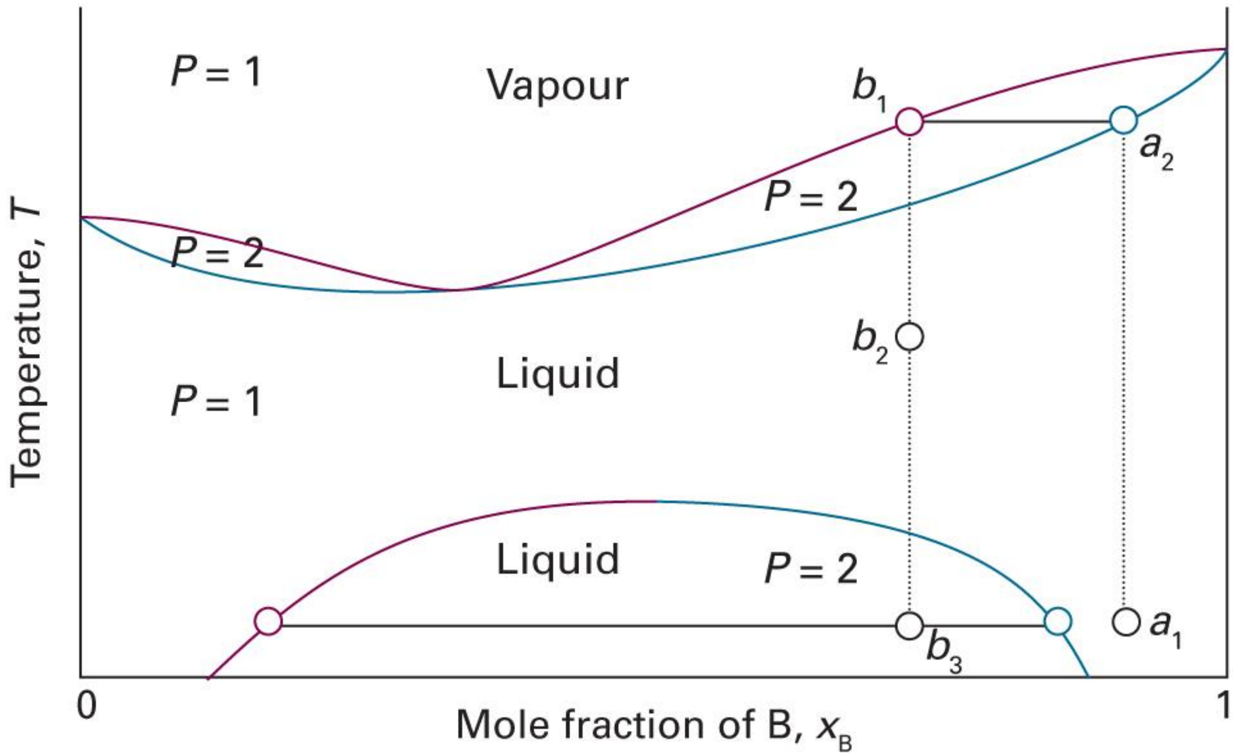




(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 12th ed., Figures 5C.12 and 5C.14)

Last class

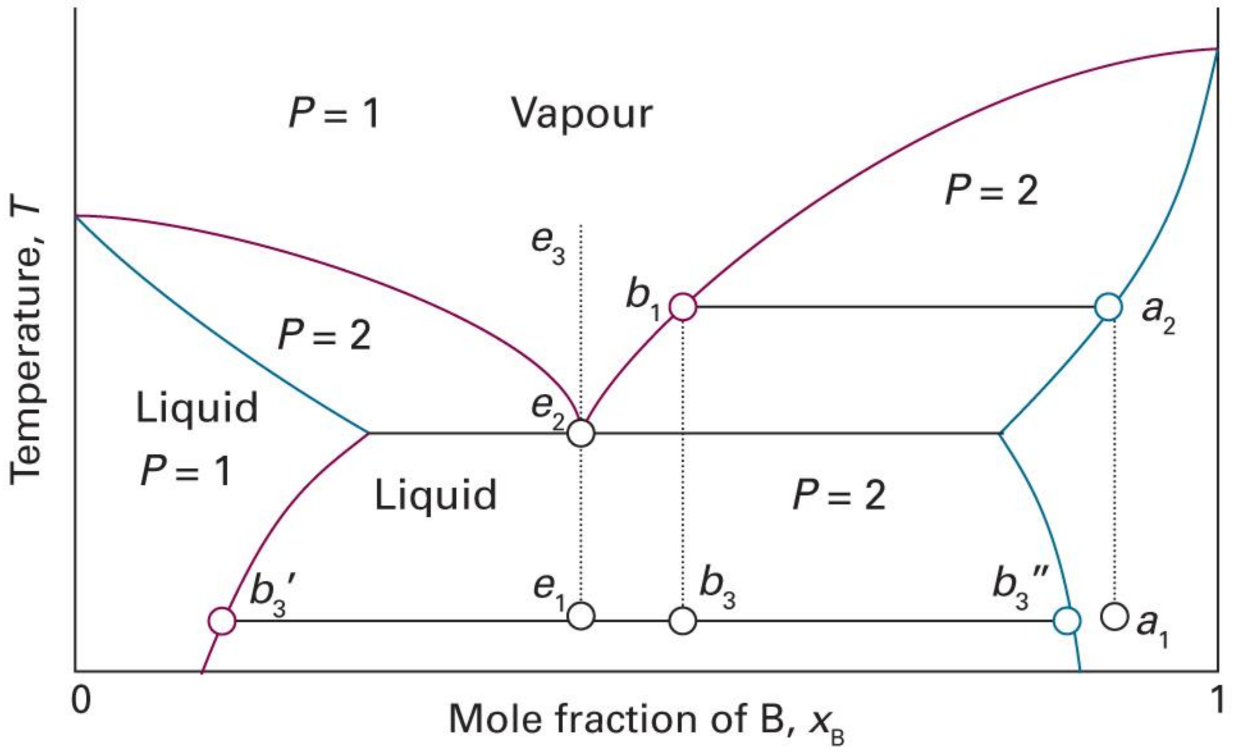
Case 1: Miscible before Boiling



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5C.21)

Last class

Case 2: Boiling before Miscible



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5C.22)

Not in Atkins

Liquid-Solid Equilibrium

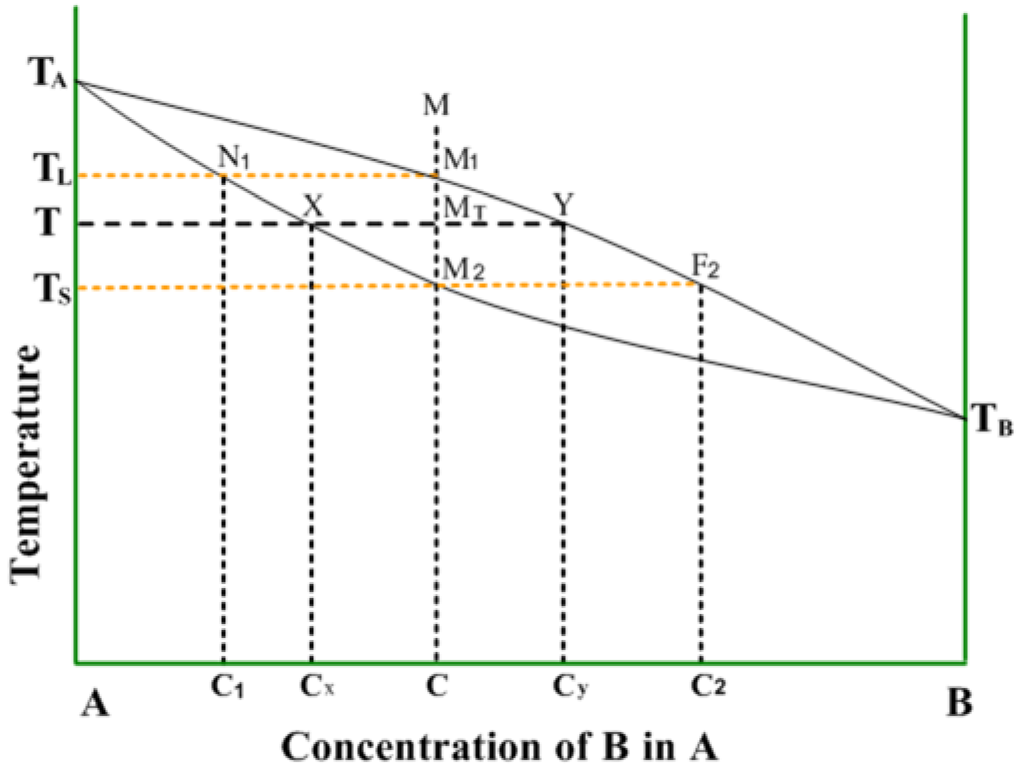
Various scenarios are possible:

1. Miscible liquid + miscible solid
2. Miscible liquid + immiscible solid
3. Miscible liquid + completely immiscible solid

We can also consider miscibility of liquid

Not in Atkins

Miscible Solid



(Image: https://www.substech.com/dokuwiki/doku.php?id=phase_transformations_and_phase_diagrams)

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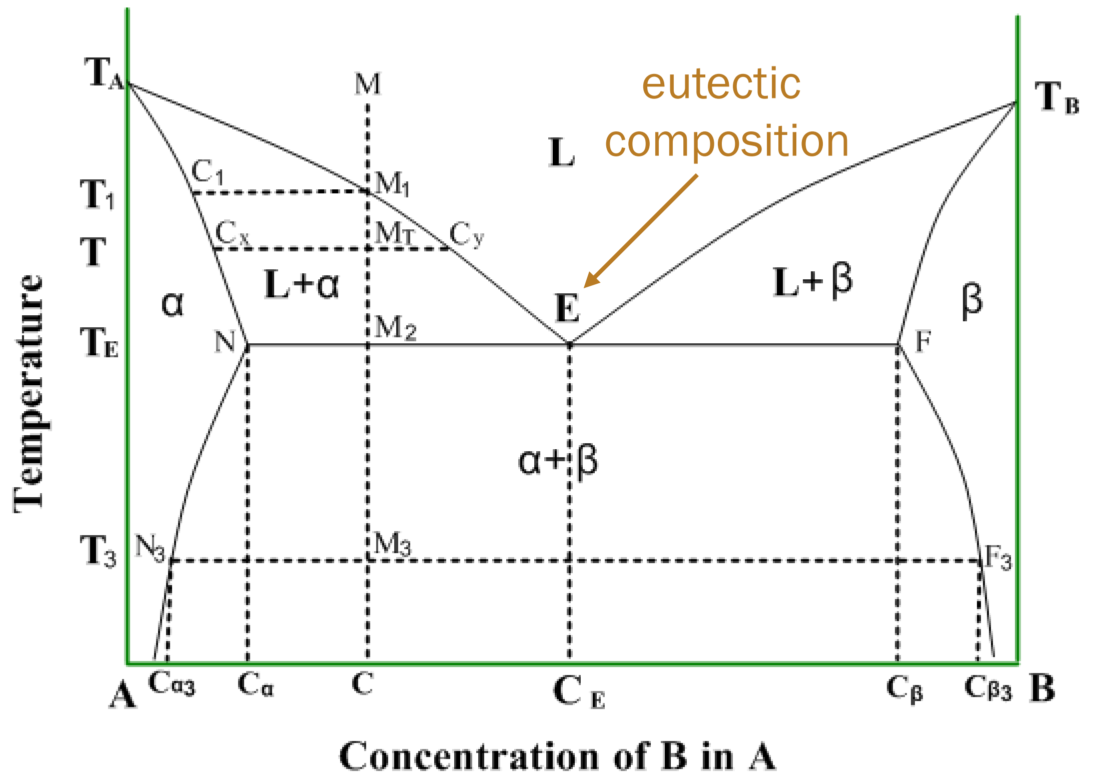
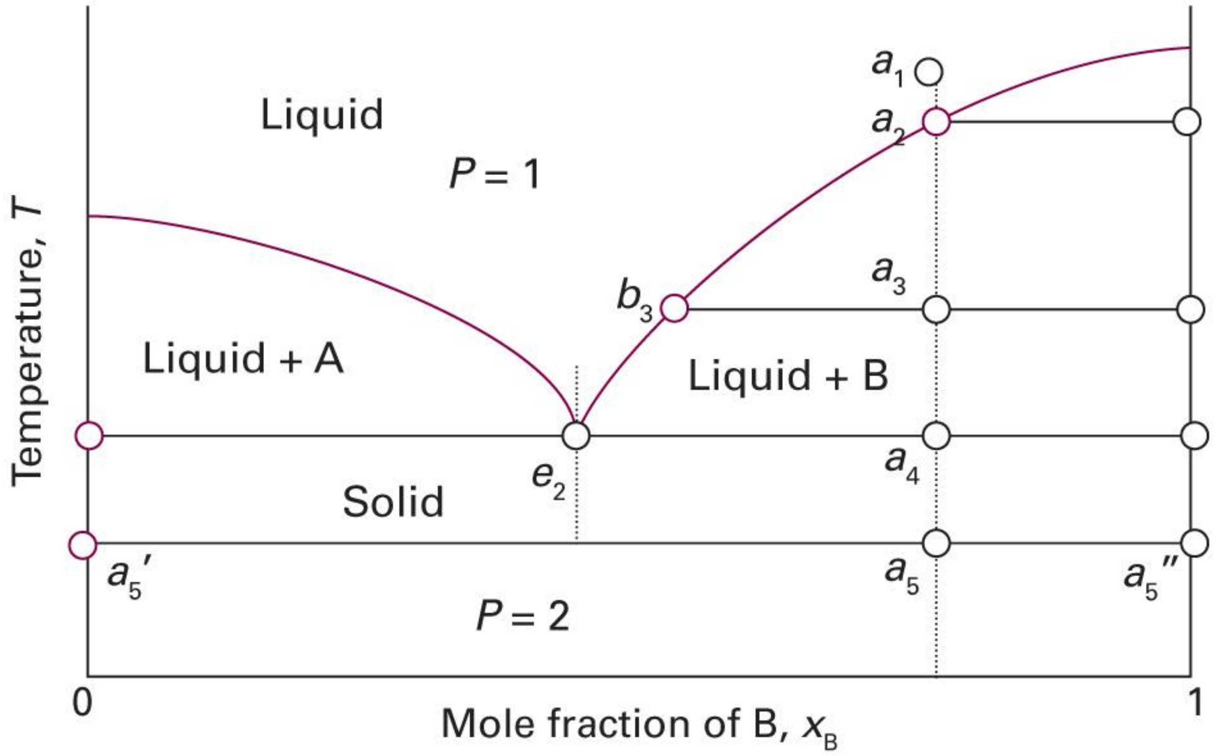
Immiscible Solid



(Image: https://www.substech.com/dokuwiki/doku.php?id=phase_transformations_and_phase_diagrams)

Topic 5D

Completely Immiscible Solid



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5D.1)

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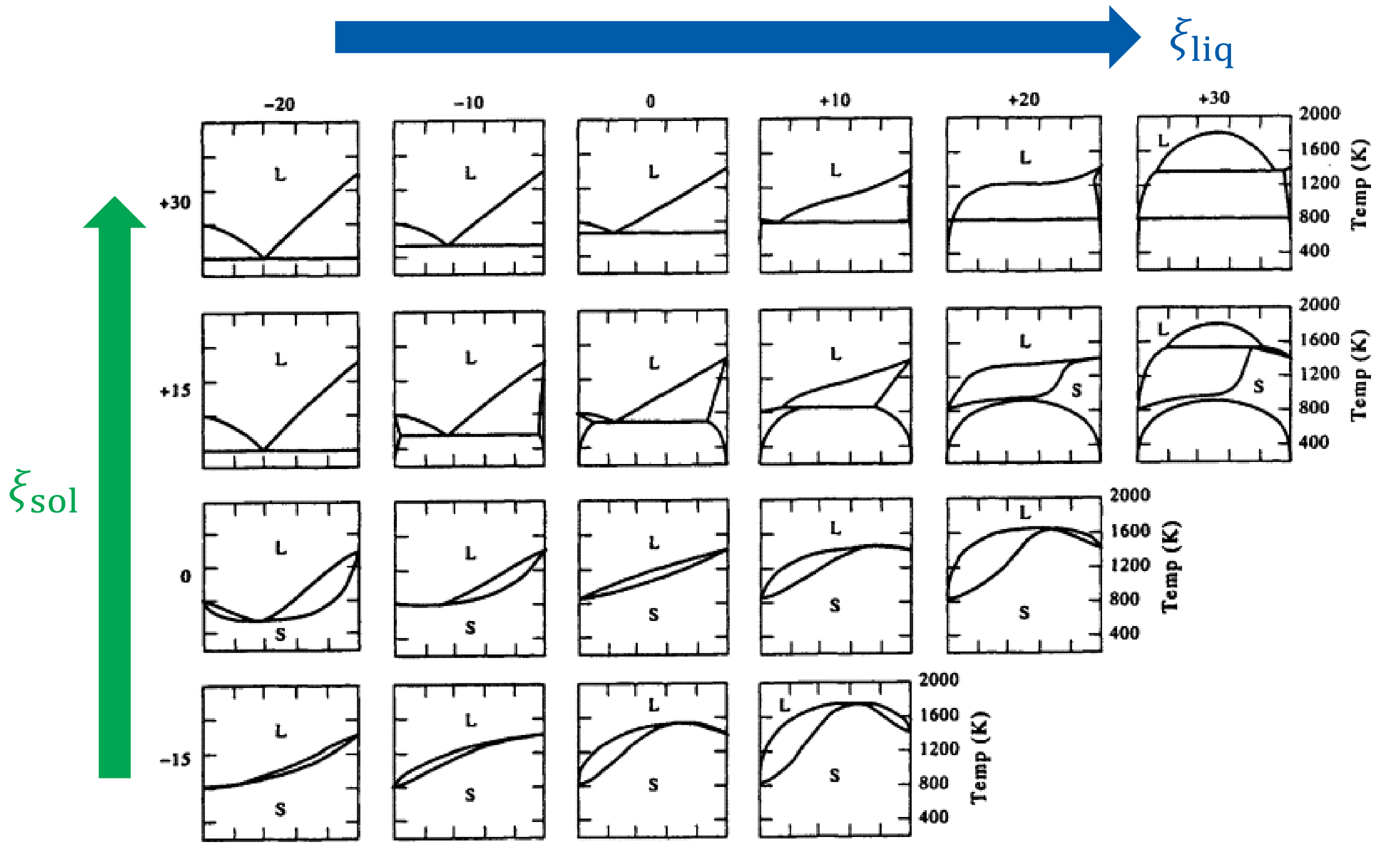
Diverse Phase Diagrams



(Image: Gaskell, *Introduction to the Thermodynamics of Materials*, 5th Ed., Fig. 10.23)

Reacting Systems

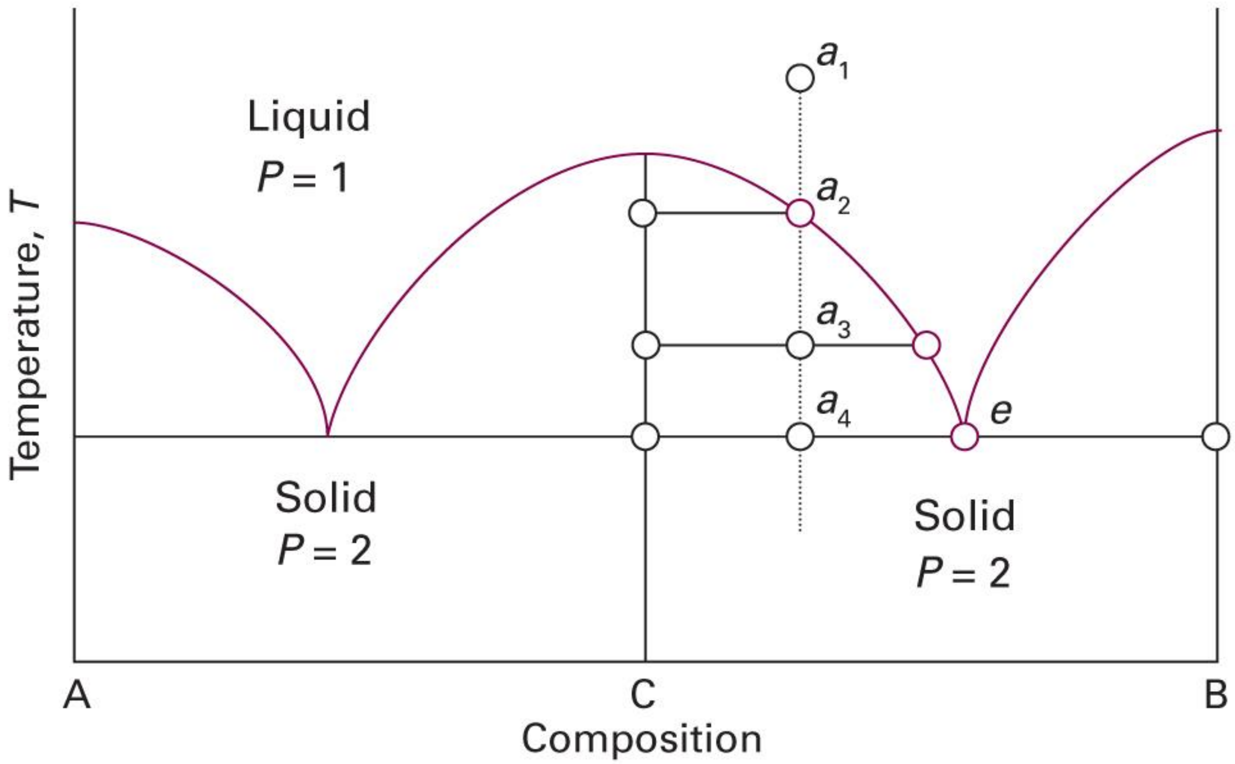
For reaction A + B $\longrightarrow$ C (irreversible),

- A 100%, B 0% = A only
- A 75%, B 25% = A 2/3, C 1/3
- A 50%, B 50% = C only
- A 25%, B 75% = B 2/3, C 1/3
- A 0%, B 100% = B only

The x-axis can be (A $\longrightarrow$ C $\longrightarrow$ B)

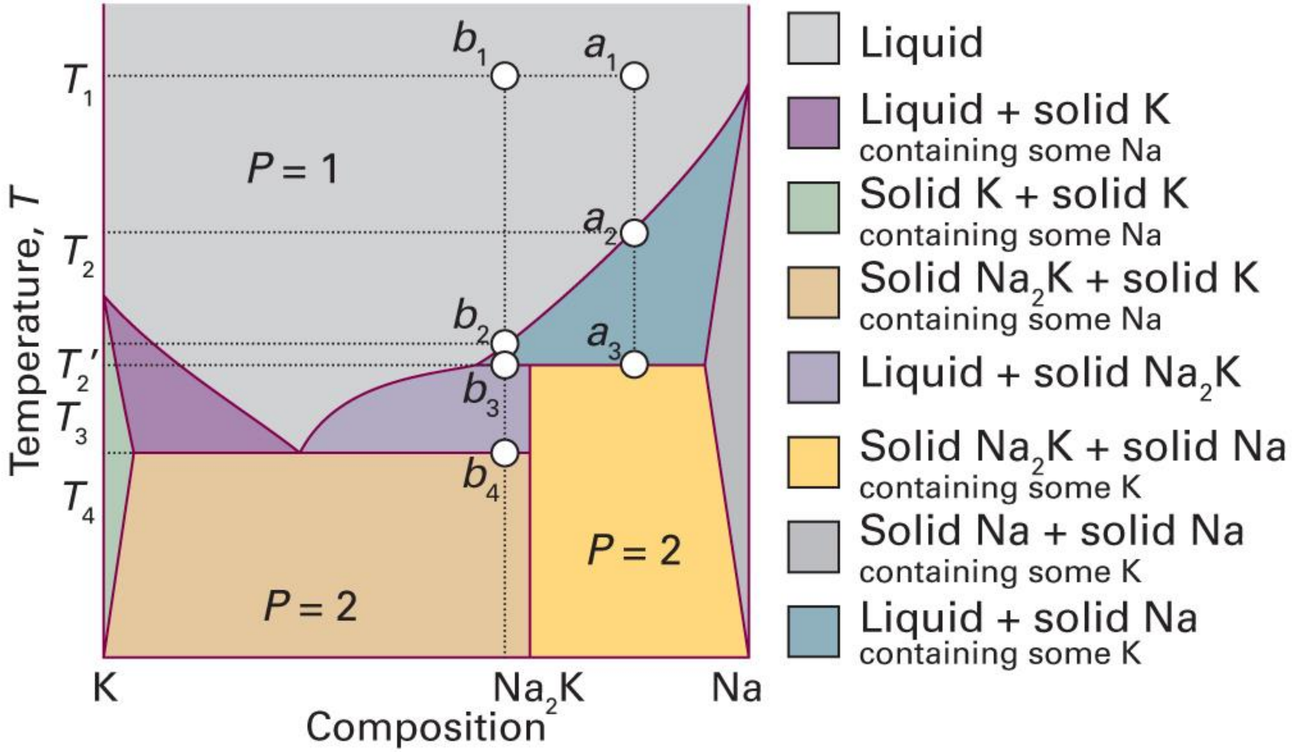
Congruent Melting

(assuming completely immiscible solid)



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5D.4)

Incongruent Melting



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5D.5)

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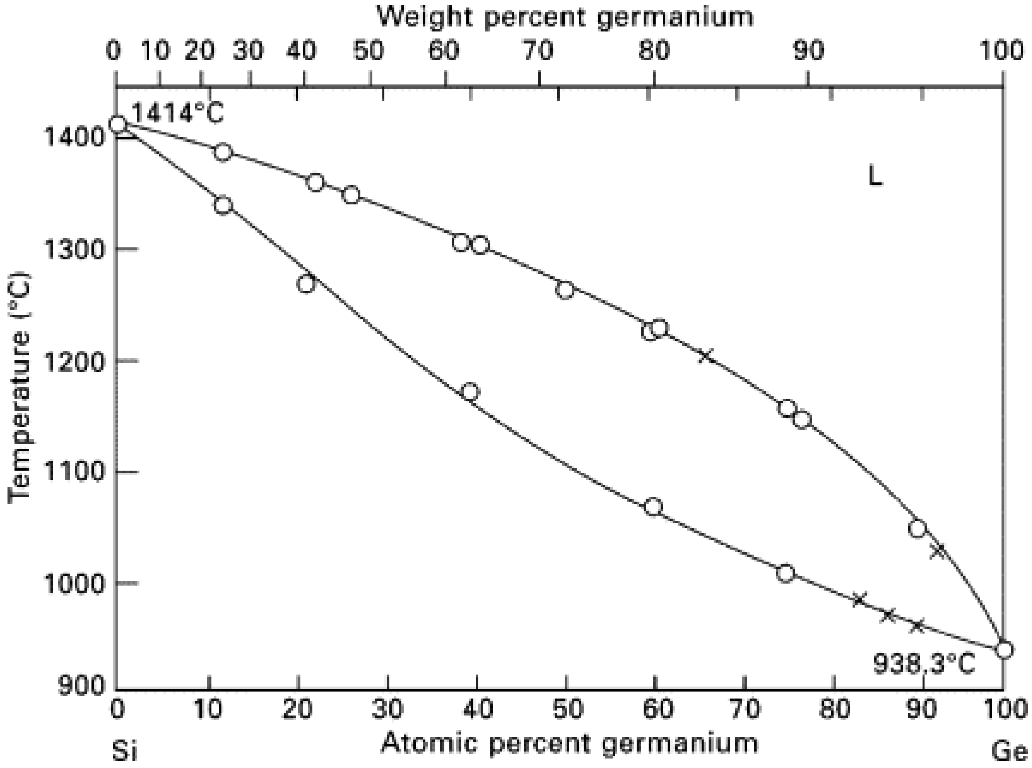
Si-Ge System



(Image: Kasper and Herzog, ***Silicon–Germanium (SiGe) Nanostructures*** (2011), Fig. 1.2)

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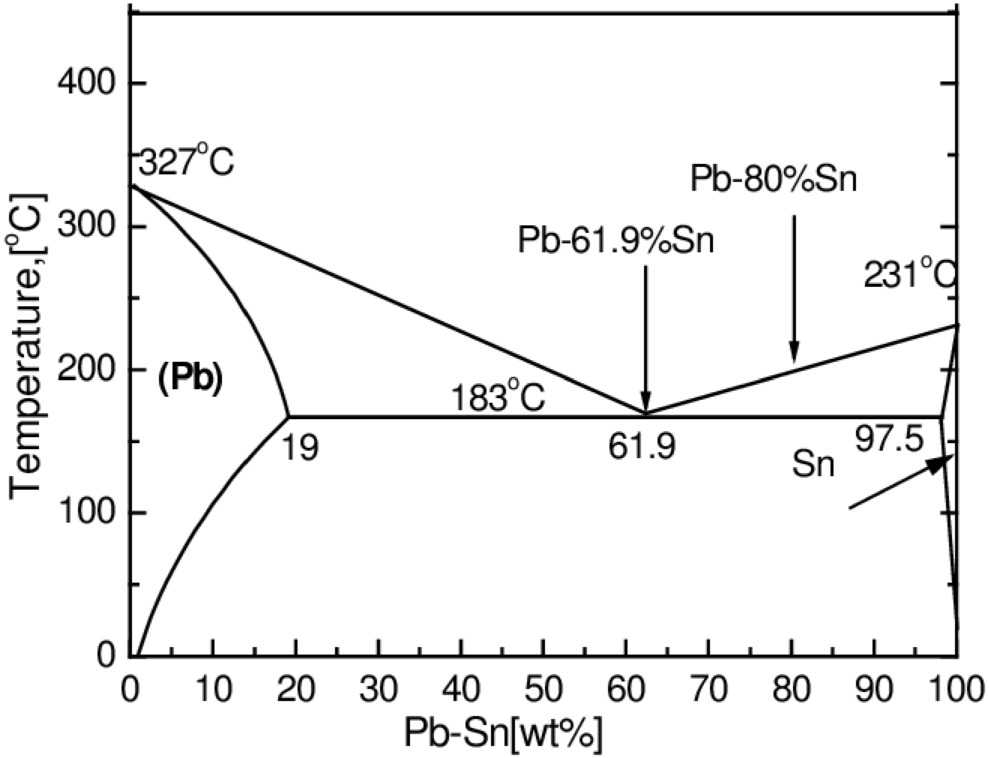
Pb-Sn System



(Image: https://www.researchgate.net/figure/The-Pb-Sn-phase-diagram_fig1_291162603)

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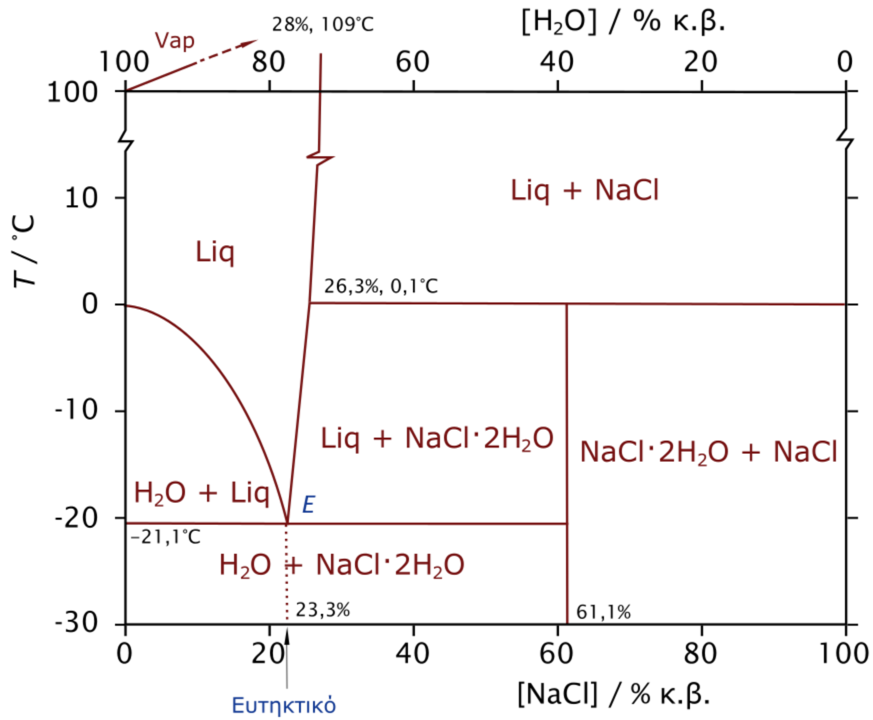
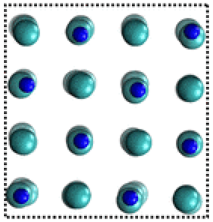
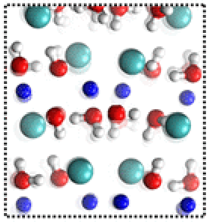
H_2O-NaCl System

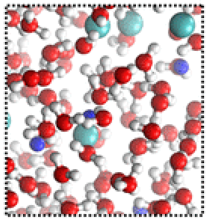




NaCl



NaCl · $2H_2O$



Liq

(Image: https://commons.wikimedia.org/wiki/File:H2O-NaCl-phase-diagram-greek.svg)

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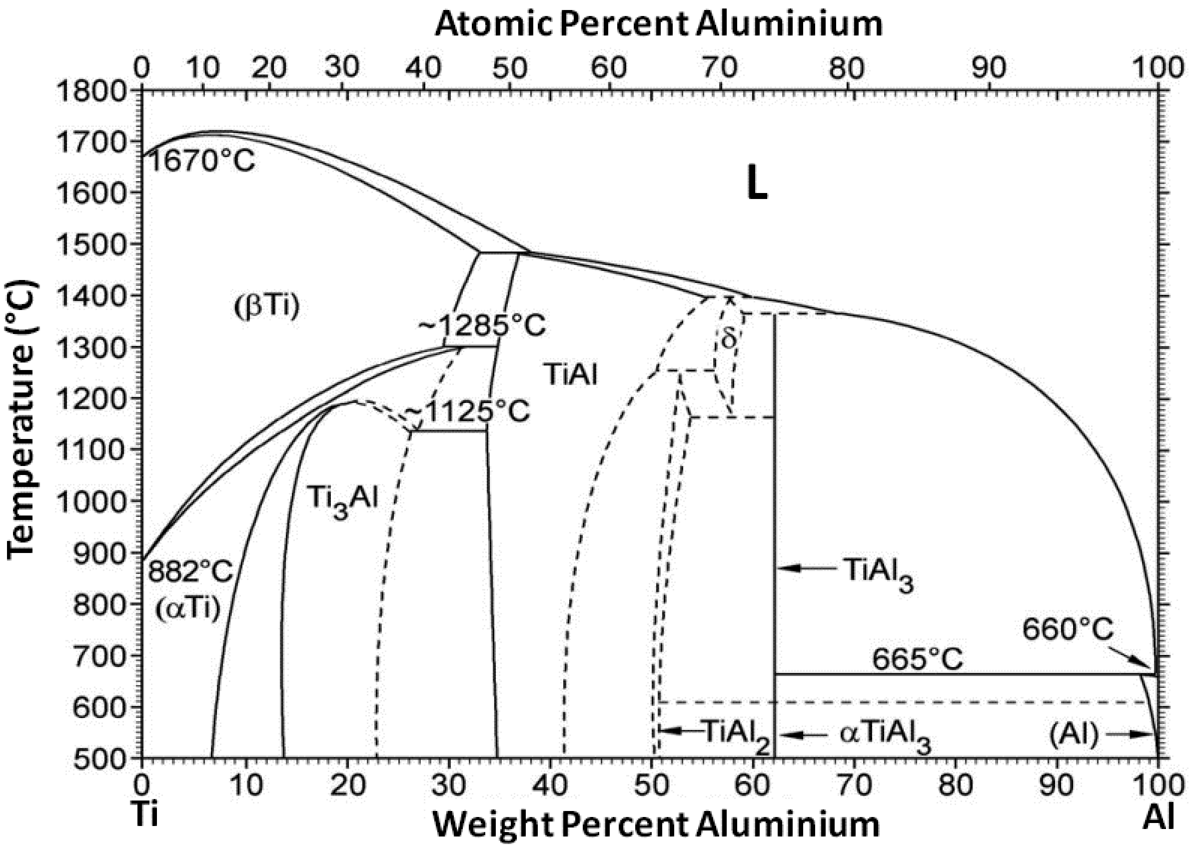
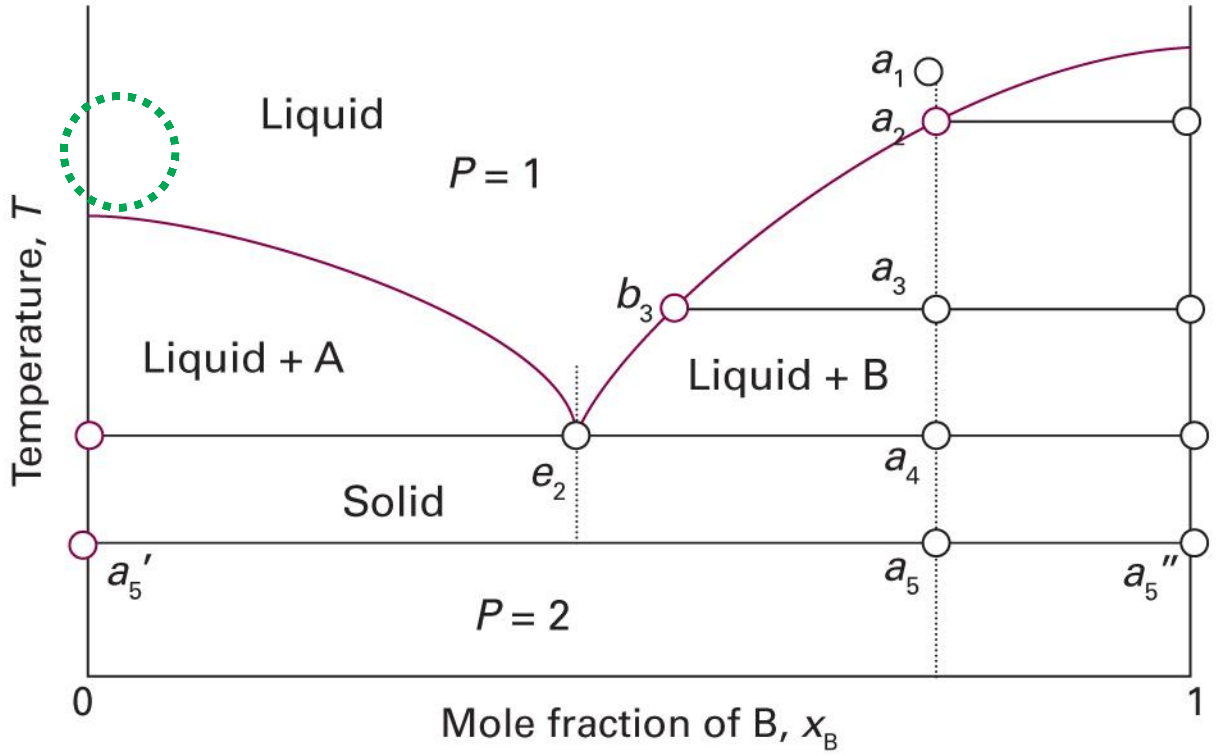
Ti-Al System



(Image: Mitra and Wanhill, *Structural Intermetallics* (2016), Figure 10)

A Little Solid in Liquid



(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 12th ed., Figure 5D.1)

A Little Solid in Liquid

- Considered as an ideal-dilute solution
  - Liquid: solvent
  - Solid: solute

- Two assumptions
  - The solute is not volatile
  - The solute does not dissolve in the solid solvent

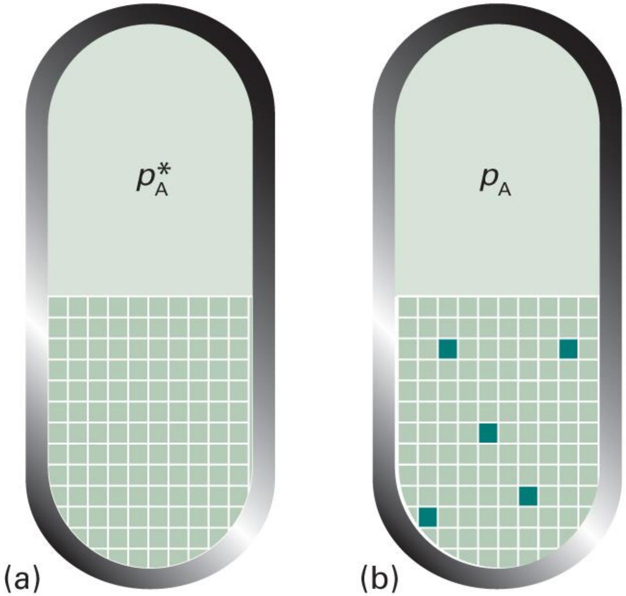
Colligative Properties

- Physical properties that depends on the relative number of solute particles present but not their chemical identity

1. Lowering of vapor pressure
2. Elevation of boiling point
3. Depression of freezing point
4. Osmotic pressure

Origin of Colligative Properties

- Due to the reduction of the chemical potential of the liquid solvent as a result of the presence of solute



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5B.7)

Not in Atkins

Lowering of Vapor Pressure

From Raoult's law,

$$p_\mathrm{A} = x_\mathrm{A} p_\mathrm{A}^*$$

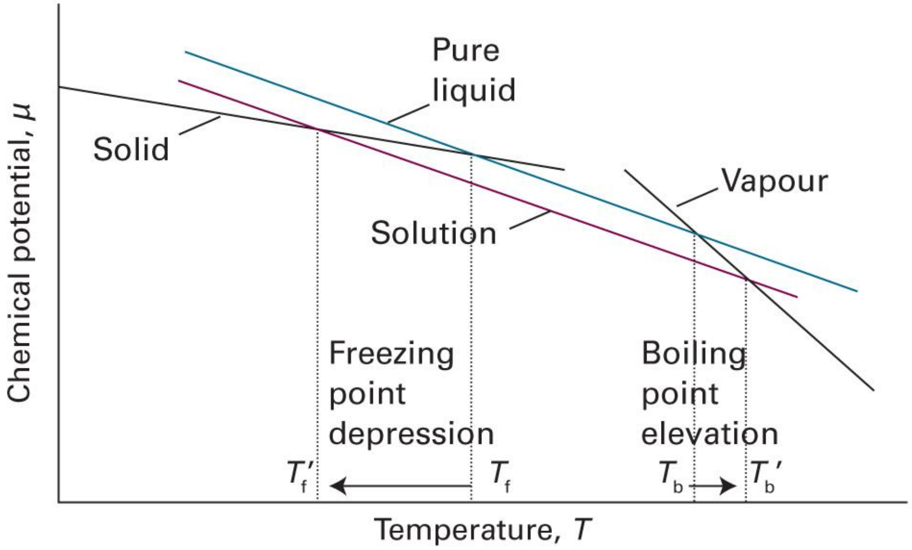
The change in vapor pressure is

$$\Delta p_\mathrm{A} = p_\mathrm{A}^* - p_\mathrm{A} = (1 - x_\mathrm{A}) p_\mathrm{A}^* = x_\mathrm{B} p_\mathrm{A}^*$$

Topic 5B

Changes in Phase Transitions

$$\mu_A = \mu_A^* + RT \ln x_A \ (x_A < 1)$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5B.6)

Boiling: Liquid-Gas Equilibrium

$$\mu_A^*(g) = \mu_A(l)$$

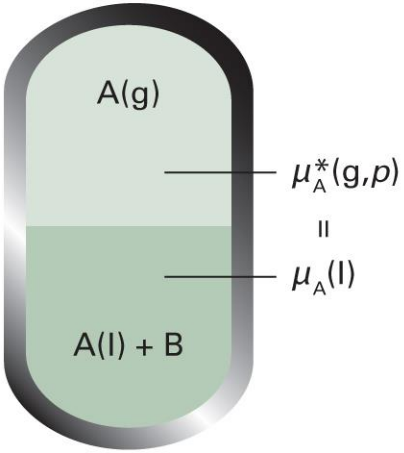
$$\mu_A^*(g) = \mu_A^*(l) + RT \ln x_A$$

which can be rearranged as

$$\ln x_A = \frac{\mu_A^*(g) - \mu_A^*(l)}{RT} = \frac{\Delta_{vap}G}{RT}$$

Using the Gibbs-Helmholtz equation,

$$\frac{d \ln x_A}{dT} = \frac{1}{R}\frac{d\left(\Delta_{vap}G/T\right)}{dT} = -\frac{\Delta_{vap}H}{RT^2}$$



(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 12th ed., Figure 5B.8)

Integration

$$\frac{\mathrm{d}\ln x_\mathrm{A}}{\mathrm{d}T} = -\frac{\Delta_\mathrm{vap}H}{RT^2}$$

$$\mathrm{d}\ln x_\mathrm{A} = -\frac{\Delta_\mathrm{vap}H}{RT^2}\mathrm{d}T$$

$$\int_\text{pure solvent}^\text{current state} \mathrm{d}\ln x_\mathrm{A} = -\frac{1}{R}\int_\text{pure solvent}^\text{current state} \frac{\Delta_\mathrm{vap}H}{T^2}\mathrm{d}T$$

$$\int_0^{\ln x_\mathrm{A}} \mathrm{d}\ln x'_\mathrm{A} = -\frac{1}{R}\int_{T^*}^{T} \frac{\Delta_\mathrm{vap}H}{(T')^2}\mathrm{d}T'$$

Evaluation

$$\int_0^{\ln x_\mathrm{A}} \mathrm{d}\ln x'_\mathrm{A} = -\frac{1}{R}\int_{T^*}^{T} \frac{\Delta_\mathrm{vap}H}{(T')^2}\mathrm{d}T'$$

Assuming that $\Delta_\mathrm{vap}H$ is constant,

$$\ln x_\mathrm{A} = -\frac{\Delta_\mathrm{vap}H}{R}\int_{T^*}^{T}\frac{1}{(T')^2}\mathrm{d}T' = -\frac{\Delta_\mathrm{vap}H}{R}\left(-\frac{1}{T}+\frac{1}{T^*}\right)$$

$$\ln(1-x_\mathrm{B}) = \frac{\Delta_\mathrm{vap}H}{R}\left(\frac{1}{T}-\frac{1}{T^*}\right)$$

Two (Valid) Approximations

$$\ln(1 - x_\mathrm{B}) = \frac{\Delta_\mathrm{vap}H}{R}\left(\frac{1}{T} - \frac{1}{T^*}\right)$$

For $x_\mathrm{B} \ll 1$, $\ln(1 - x_\mathrm{B}) \approx -x_\mathrm{B}$ and

$$x_\mathrm{B} \approx \frac{\Delta_\mathrm{vap}H}{R}\left(\frac{1}{T^*} - \frac{1}{T}\right) = \frac{\Delta_\mathrm{vap}H}{RTT^*}(T - T^*)$$

Also, supposing $\Delta T_\mathrm{b} = T - T^* \ll T^*$,

$$x_\mathrm{B} \approx \frac{\Delta_\mathrm{vap}H}{R(T^*)^2}\Delta T_\mathrm{b}$$

Elevation of Boiling Point

$$x_{\mathrm{B}} \approx \frac{\Delta_{\mathrm{vap}} H}{R(T^*)^2} \Delta T_{\mathrm{b}}$$

Rearrangement gives

$$\Delta T_{\mathrm{b}} = K x_{\mathrm{B}}$$

where $K = R(T^*)^2/\Delta_{\mathrm{vap}}H$.

Using the molality,

$$\Delta T_{\mathrm{b}} = K_{\mathrm{b}} b$$

where K_{b} is called the boiling-point constant.

Freezing: Solid-Liquid Equilibrium

$$\mu_{\mathrm{A}}^{*}(s) = \mu_{\mathrm{A}}(l)$$

$$\mu_{\mathrm{A}}^{*}(s) = \mu_{\mathrm{A}}^{*}(l) + RT\ln x_{\mathrm{A}}$$

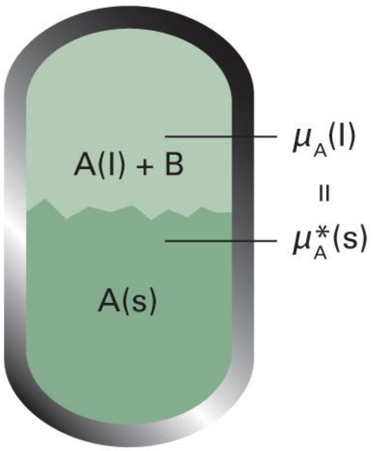
Similarly, we obtain

$$\Delta T_{\mathrm{f}} = K' x_{\mathrm{B}}$$

where $K' = R(T^{*})^{2}/\Delta_{\mathrm{fus}}H$, or,

$$\Delta T_{\mathrm{f}} = K_{\mathrm{f}} b$$

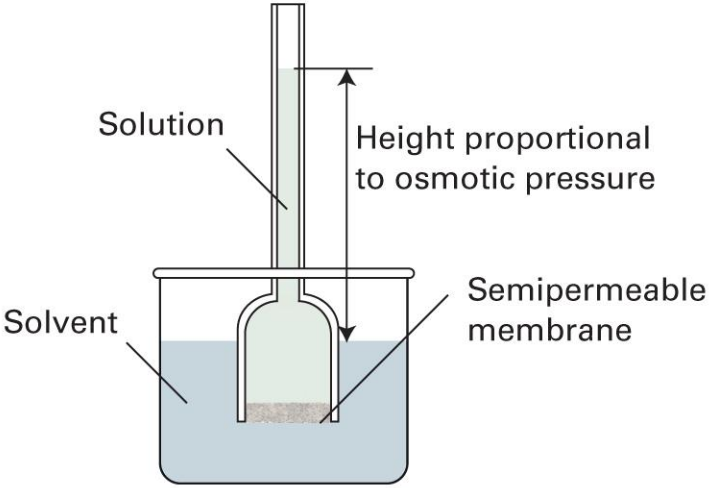
where K_{f} is called the freezing-point constant.



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5B.9)

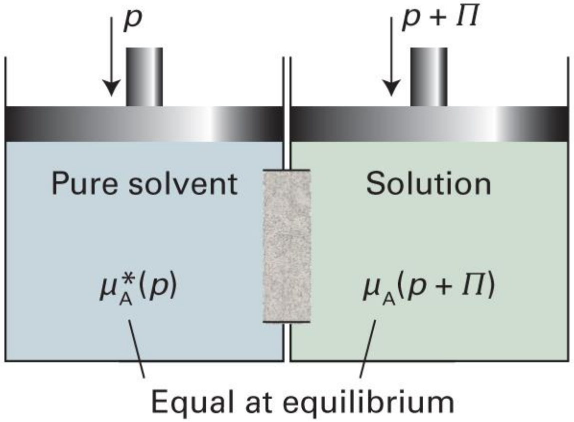
Osmosis

the spontaneous passage of a pure solvent into a solution separated from it by a semipermeable membrane



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5B.13)

Thermodynamics of Osmosis



$$\mu_A^*(p) = \mu_A(x_A, p + \Pi)$$

$$\mu_A^*(p) = \mu_A^*(p + \Pi) + RT \ln x_A$$

Thus,

$$\mu_A^*(p + \Pi) = \mu_A^*(p) - RT \ln x_A$$

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5B.12)

Pressure Effect

In general,

$$G_{\mathrm{m}}(p_2) = G_{\mathrm{m}}(p_1) + \int_{p_1}^{p_2} V_{\mathrm{m}}\,\mathrm{d}p$$

$$\mu_{\mathrm{A}}^{*}(p+\Pi) = \mu_{\mathrm{A}}^{*}(p) + \int_{p}^{p+\Pi} V_{\mathrm{m}}\,\mathrm{d}p'$$

Assuming that V_{m} is constant,

$$\mu_{\mathrm{A}}^{*}(p+\Pi) = \mu_{\mathrm{A}}^{*}(p) + V_{\mathrm{m}} \int_{p}^{p+\Pi} 1\,\mathrm{d}p' = \mu_{\mathrm{A}}^{*}(p) + V_{\mathrm{m}}\Pi$$

Combining the Two

$$\mu_A^*(p+\Pi) = \mu_A^*(p) - RT\ln x_A$$

$$\mu_A^*(p+\Pi) = \mu_A^*(p) + V_m\Pi$$

Comparing the two equations,

$$-RT\ln x_A = V_m\Pi$$

$$-RT\ln(1-x_B) = V_m\Pi$$

Since $\ln(1-x_B) \approx -x_B$ for $x_B \ll 1$,

$$RTx_B = V_m\Pi$$

Osmotic Pressure

$$RTx_B = V_m\Pi$$

For a dilute solution,

$$x_B = n_B/(n_A + n_B) \approx n_B/n_A$$

Thus,

$$RTn_B = n_A V_m \Pi = V\Pi$$

and

$\Pi = [B]RT$ (van’t Hoff equation)

Real Solutions

For an ideal solution (now solute = J),

$$\Pi = [\mathrm{J}]RT$$

The osmotic virial expansion is

$$\Pi = [\mathrm{J}]RT(1 + B[\mathrm{J}] + C[\mathrm{J}]^2 + \cdots)$$

where $B, C, \ldots$ are the osmotic virial coefficients.

Second Coefficient

$$\Pi = [\mathrm{J}]RT(1 + B[\mathrm{J}] + \cdots)$$

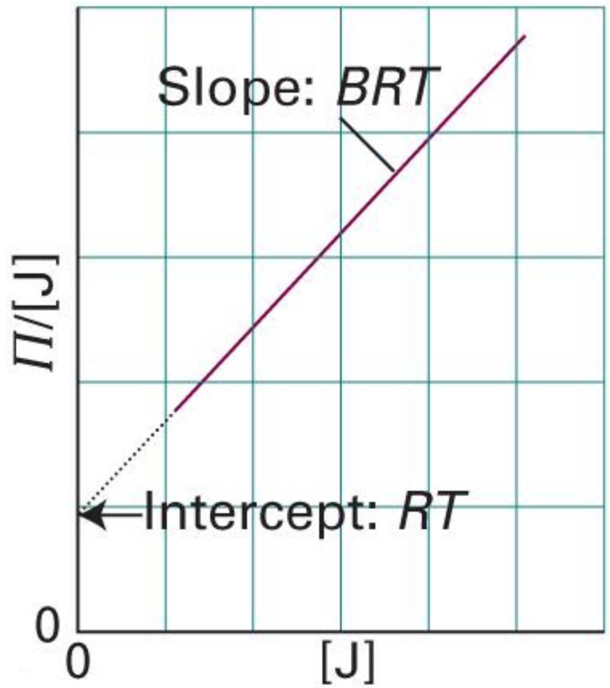
Taking the first correction term only,

$$\Pi \approx [\mathrm{J}]RT(1 + B[\mathrm{J}])$$

- $B > 0$: solute-solute repulsion dominant
- $B \approx 0$: similar to ideal-dilute solution
- $B < 0$: solute-solute attraction dominant

Osmometry

$$\Pi/[\mathrm{J}] \approx RT + BRT[\mathrm{J}]$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5B.14)

Not in Atkins

Case of Ionic Solutes

e.g. $NaCl(s) \longrightarrow Na^+(aq) + Cl^-(aq)$

- van't Hoff factor i

$$\Delta T_\mathrm{b} = iK_\mathrm{b}b$$

$$\Delta T_\mathrm{f} = iK_\mathrm{f}b$$

$$\Pi = i[\mathrm{B}]RT$$

Not in Atkins

van't Hoff Factors

0.05000 M, 25℃

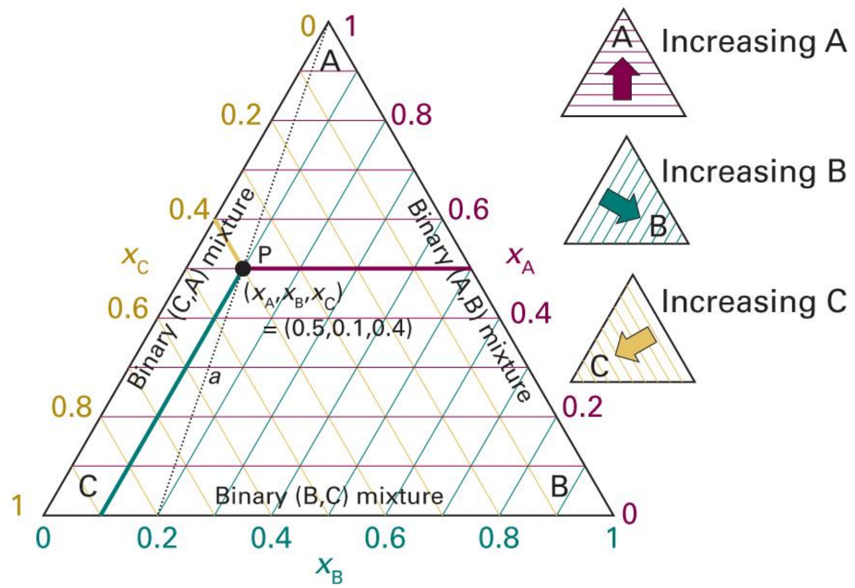
Compound	*i* (ideal)	*i* (measured)
glucose	1	1.0
NaCl	2	1.9
KCl	2	1.9
$MgCl_2$	3	2.7
$FeCl_3$	4	3.4
$Ca(NO_3)_2$	3	2.5
$AlCl_3$	4	3.2
$MgSO_4$	2	1.4

(Image: https://chem.libretexts.org/@go/page/24261)

Topic 5E

Ternary Systems: Triangle

$$x_{\mathrm{A}} + x_{\mathrm{B}} + x_{\mathrm{C}} = 1$$

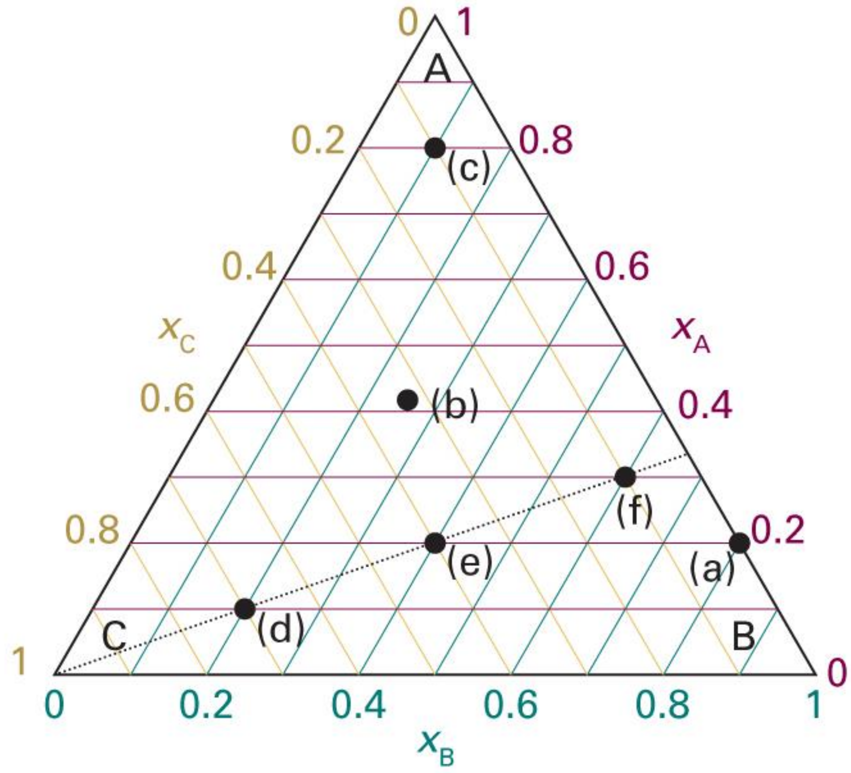


(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5E.1)

Brief Illustration 5E.1

Indicate the following points:

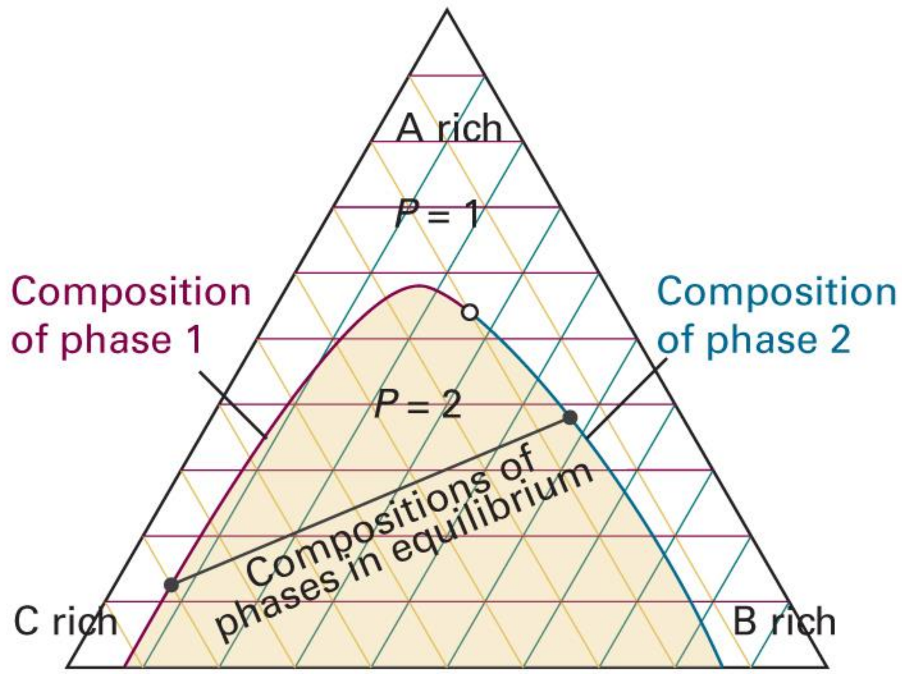
Point	x_A	x_B	x_C
a	0.20	0.80	0
b	0.42	0.26	0.32
c	0.80	0.10	0.10
d	0.10	0.20	0.70
e	0.20	0.40	0.40
f	0.30	0.60	0.10



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5E.2)

Topic 5E

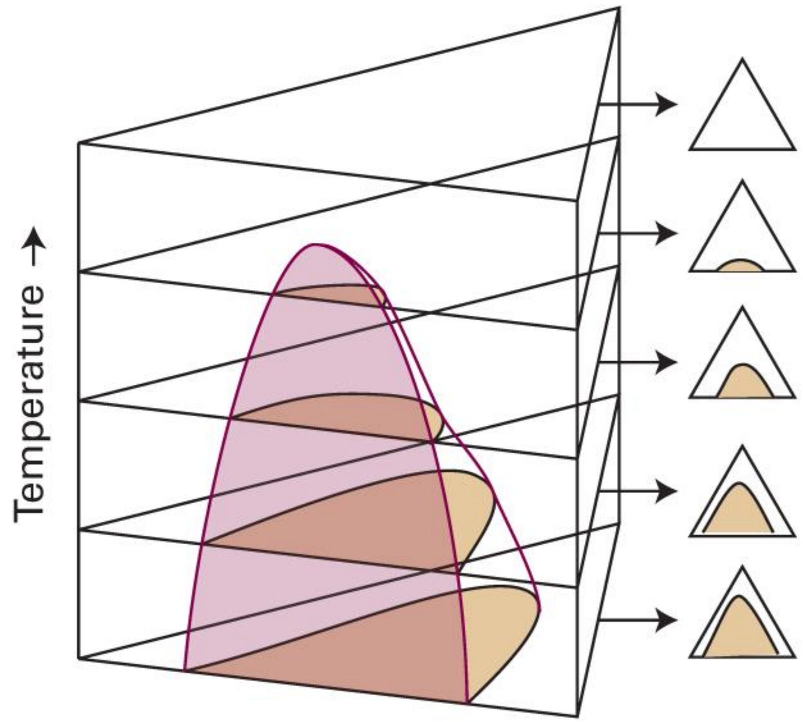
Partially Miscible Liquids



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5E.5)

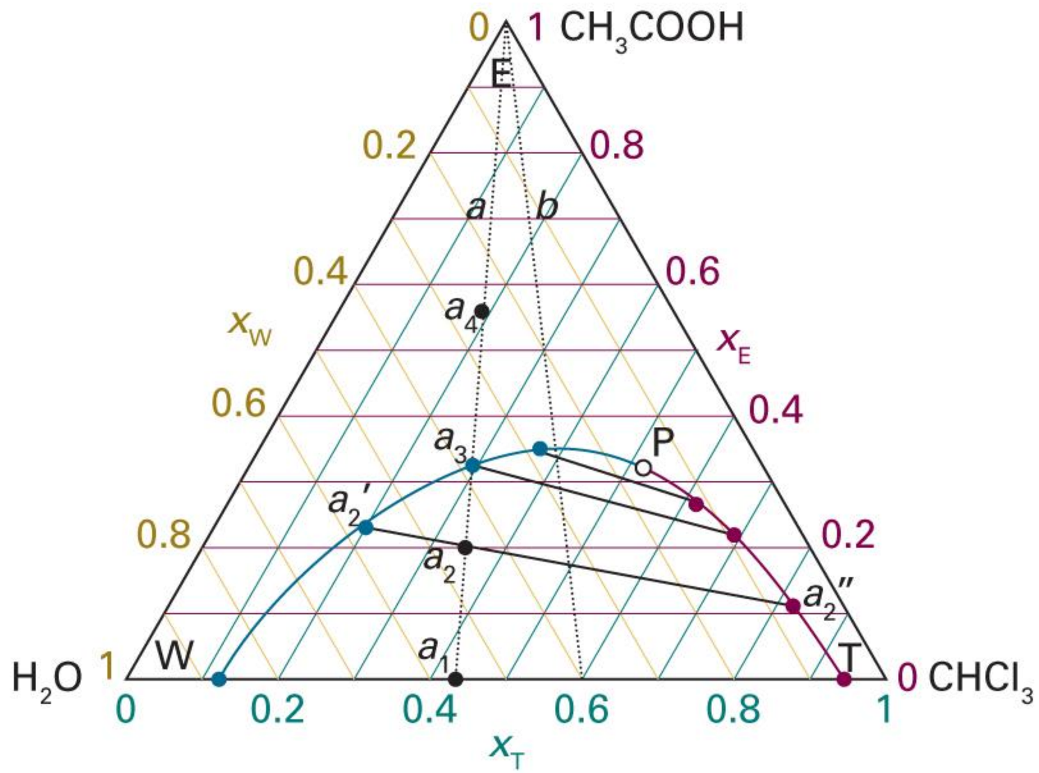
Topic 5E

What If Temperature Changes?



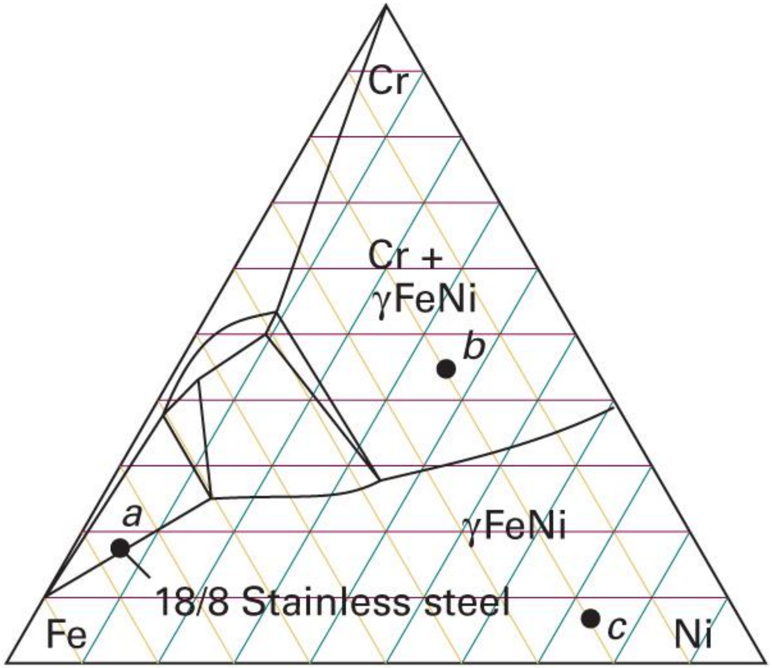
(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5E.3)

Example: CH_3COOH-H_2O-$CHCl_3$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5E.4)

Ternary Solids



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 5E.6)

Summary

- Phase
- Chemical potential
- Gibbs-Duhem eq.

Multi-component system

Gas mixture

- Dalton's law
- Perfect gas
- Real gas

Ternary systems

Liquid mixture

Ideal solution

- Raoult's law

$$p_A = x_A p_A^*$$

- Chemical potential

$$\mu_A = \mu_A^* + RT \ln x_A$$

- $\Delta_{mix}G = -T\Delta_{mix}S$

Ideal-dilute solution

- Henry's law

$$p_B = x_B K_B$$

- Chemical potential

$$\mu_B = \mu_B^{\ominus} + RT \ln x_B$$

Real solution

- Regular solution
- Bragg-Williams model
- Excess function
- Activity

$$\mu_A = \mu_A^* + RT \ln a_A$$

$$\mu_B = \mu_B^{\ominus} + RT \ln a_B$$

- L-G phase diagram
- L-L phase diagram
- L-S phase diagram

Colligative properties